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To cite this article: Hani H. Al-Baadani, Abdulrahman S. Alharthi, Nasif I. Abbas, Akram A. Qasem, Ali Saleh & Mohamed A. Ibraheem (2025) Effect of activated and inactivated *Saccharomyces cerevisiae* as alternative to antibiotic growth promoter on the performance and health of broilers infected with *Clostridium perfringens*, Italian Journal of Animal Science, 24:1, 43-52, DOI: [10.1080/1828051X.2024.2441349](https://doi.org/10.1080/1828051X.2024.2441349)

To link to this article: <https://doi.org/10.1080/1828051X.2024.2441349>



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Published online: 20 Dec 2024.



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RESEARCH ARTICLE



## Effect of activated and inactivated *Saccharomyces cerevisiae* as alternative to antibiotic growth promoter on the performance and health of broilers infected with *Clostridium perfringens*

Hani H. Al-Baadani<sup>a</sup>, Abdulrahman S. Alharthi<sup>a</sup>, Nasif I. Abbas<sup>a</sup>, Akram A. Qasem<sup>b</sup>, Ali Saleh<sup>b</sup> and Mohamed A. Ibraheem<sup>b</sup>

<sup>a</sup>Department of Animal Production, King Saud University, Riyadh, Saudi Arabia; <sup>b</sup>Department of Food Science and Nutrition, King Saud University, Riyadh, Saudi Arabia

### ABSTRACT

The study aims to evaluate the addition of activated and inactivated *Saccharomyces cerevisiae* on the performance and health of broilers infected with *Clostridium perfringens*. 360 1-day-old male Ross 308 broilers were divided into 5 groups of 12 cages as experimental units (6 birds per cage) as follows: NC = negative control, basal diet; PC = positive control, basal diet; T1 = 0.5 g amoxicillin 20%/kg diet; T2 = 0.75 g activated *Saccharomyces cerevisiae*/kg diet; T3 = 0.75 g inactivated *Saccharomyces cerevisiae*/kg diet. At 20 and 21 days of age, all groups except NC were infected with *Clostridium perfringens* type A by oral injection. At 30 days of age, carcass traits, liver enzymes, oxidation status, microbiota and ileum histomorphology were evaluated. The performance indicators, carcass weight after infection of the birds, were negatively affected compared to the NC group, while T2 and T3, followed by T1, improved compared to the PC group ( $p < .05$ ). The concentrations of ALT, AST, ALP, and MDA as well as *Clostridium spp* and *Salmonella spp*. increased in the PC group, and the TAC concentration and *Lactobacillus spp*. decreased in the PC group, while these parameters improved by T2 and T3 ( $p < .05$ ). In addition, the PC group had the lowest histomorphometric parameters compared to NC, while they increased in other groups than PC ( $p < .05$ ). In conclusion, activated and inactivated *Saccharomyces cerevisiae* could be effectively mitigate the negative effects of *Clostridium perfringens* infection on broiler performance and health.

### HIGHLIGHTS

- Infection with *Clostridium perfringens* has a negative effect on broiler performance, carcass weight, liver enzymes, oxidative status, microbiota and ileal histomorphology.
- Supplementation with activated and inactivated *Saccharomyces cerevisiae* mitigates these negative effects.
- Supplementation improves performance and intestinal health, reduces oxidative stress and modulates the gut microbiota with reduces *Clostridium perfringens* colonisation.

### ARTICLE HISTORY

Received 7 November 2024  
Revised 27 November 2024  
Accepted 7 December 2024

### KEYWORDS

*S. cerevisiae*; necrotic enteritis; performance; health; broilers

## Introduction

The poultry industry, particularly broiler chickens, faces several challenges, including disease outbreaks, suboptimal nutrition, and antibiotic resistance that can significantly impact on bird health, productivity and overall economic viability (Nawab et al. 2018). One of the most pressing issues is the spread of bacterial infections such as *Clostridium perfringens* (*C. perfringens*), which can lead to necrotic enteritis, a devastating disease that causes severe intestinal damage, reduced feed efficiency, and increased mortality (Moore 2016). However, the overuse of antibiotic growth promoters in the production of broiler chickens

has resulted in the rise of resistant strains to the drugs such as *C. perfringens*, making the disease more difficult to treat (Zheng et al. 2021). To mitigate the negative effects of these pathogens, researchers and producers have explored various strategies, including the use of dietary supplements such as probiotics and prebiotics (Ahmed et al. 2023a; Fornazier et al. 2024). Probiotics are beneficial microorganisms that play an important role in host health benefits by modulating the gut microbiota, improving digestion and enhance immunity (Sánchez et al. 2017). Prebiotics are indigestible feed components that selectively stimulate the growth and activity of helpful

**CONTACT** Hani H. Al-Baadani  [hsaheed@ksu.edu.sa](mailto:hsaheed@ksu.edu.sa)

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commensal bacteria in the caecum (Ayyash and Liu 2023). By providing a feed source for these beneficial bacteria, prebiotics can help improve gut health and reduce pathogen colonisation (Adhikari and Kim 2017). The most commonly used probiotic or prebiotic in poultry nutrition is *Saccharomyces cerevisiae* (*S. cerevisiae*) (Dhama and Singh 2010). The main components of the cell wall of *S. cerevisiae* are  $\alpha$ -mannan (31% of dry matter),  $\beta$ -glucan (60% of dry matter) and chitin (about 2% of dry matter), with two different forms of  $\beta$ -glucan linkages:  $\beta$  1,3- and  $\beta$  1,6-glucan (Yousefi 2023). The wall of *S. cerevisiae* contains  $\beta$ -glucans and MOS, which inhibit pathogenic bacteria (Faustino et al. 2021).

One study suggests that supplementation of *S. cerevisiae* cell wall (prebiotics) (0.25 or 0.50 g/kg) leads to recovery of performance in broilers affected by necrotic enteritis (Alqhtani et al. 2024). In addition, supplementation of broiler diets with *S. cerevisiae* cell wall has been shown to positively influence broiler production performance and gut health (Pascual et al. 2020). In contrast, Askri et al. (2022) reported that a prebiotic derived from *S. cerevisiae* (1–2 g/kg feed) does not improve performance but can maintain meat quality. Dietary supplementation with live *S. cerevisiae* (probiotic) at a dosage of 0.5 and 1 g/kg increased performance, antioxidant capacity and improved intestinal morphology of broilers and could be used as a potential alternative to antibiotics (He et al. 2021). Studies have shown that activated and inactivated *S. cerevisiae* (probiotics and prebiotics) improve the gut health and performance of broiler chickens by modulating the gut microbiota, reducing pathogen colonisation and improving nutrient absorption (Lin et al. 2014; Khalid et al. 2021; Fazelnia et al. 2021). The health of broiler chickens can be affected by oxidative stress caused by an imbalance between the generation of reactive oxygen species and the antioxidant defense system (Ménégaud et al. 2023). Study showed that probiotics and prebiotics can help reduce oxidative stress in broilers (Ogwiji et al. 2024). Liver enzymes such as ALT and AST play an important role in metabolism in the liver and in many different tissues (Can and İmik 2024). Therefore, they are considered indicators of liver health (Wang et al. 2015). However, study have shown that the addition of probiotics (*S. cerevisiae*) lowers ALT and AST levels in broilers (Udeh et al. 2021). In addition, serum AST and ALT levels decreased in broiler chickens in response to the addition of prebiotics (Yalçinkaya et al. 2008).

This study may provide valuable insight to determine the optimal form of *S. cerevisiae* (activated and inactivated) as a feed additive to improve broiler

health and performance, especially in the context of *C. perfringens* infection, by reducing the severity of necrotic enteritis through modulation of the gut microbiota and possible inhibition of *C. perfringens* growth. In addition, this investigation could contribute to the development of more effective strategies for broiler production and health by comprehensively evaluating several parameters, as hypothesised in the current study. However, the aim of the present study was to investigate the effect of activated and inactivated *S. cerevisiae* as a feed additive on performance indicators, carcase traits, liver enzymes, antioxidant activity, oxidative stress, selected microbiota and ileum histomorphology in broilers infected with *C. perfringens*.

## Materials and methods

### Ethical approved

The current study was approved by the Scientific Research Ethics Committee of the King Saud University Research Council (SE-21–47) and follows the ARRIVE guidelines for animal experimentation

### Birds, husbandry and diets

Three hundred and sixty male Ross 308 broiler chicks at one day of age (initial body weight =  $42.57 \pm 0.22$  g) were divided into sixty cages (70 cm<sup>2</sup>) as the experimental units (72 birds per group, 6 birds per cage). The study was conducted for thirty days in an environmentally controlled battery room. The environmental conditions were maintained at 33 °C and 45% humidity at one day of age, gradually decreasing to 20 °C and 70% humidity at 24 d of age. A 24 h of light (35 lux) during the feeding of the starter diet birds (1–15 d) and then lowered to 18 h of light (10 lux) with 6 h of darkness during the feeding of the grower diet birds (16–30 d). Both the environmental conditions and the lighting program were conducted according to the Ross 308 breed guide (Aviagen 2018). Feed and water were provided *ad libitum* throughout the experiment. All birds were vaccinated by nasal instillation, including Newcastle disease, infectious bronchitis and infectious bursal disease (Fort Dodge Animal Health-USA) at 1 and 7 d of age. The control diets were based on yellow corn and soy-bean meal and were formulated according to the requirements of the commercial Ross 308 breed (Aviagen 2022) as shown in Table 1. The five diet groups were divided using completely randomised design as follows: NC = negative control, birds on control diet + uninfected; PC = positive control, birds on control diet + infected; TI = birds on 0.50 g Amoxicillin

**Table 1.** Ingredients and nutrients calculated for the basal feed.

| Ingredients (g/kg)          | Feeding periods |         |
|-----------------------------|-----------------|---------|
|                             | 1–15 d          | 16–30 d |
| Corn                        | 625.70          | 670.00  |
| Soybean meal (48% CP)       | 310.00          | 264.30  |
| Corn oil                    | 21.90           | 28.00   |
| L-lysine                    | 1.80            | 2.40    |
| DL-methionine               | 2.60            | 2.00    |
| Threonine                   | 0.70            | 2.50    |
| Ground limestone            | 7.30            | 5.90    |
| Di-calcium phosphate        | 25.00           | 20.00   |
| Salt                        | 2.50            | 2.50    |
| Choline chloride            | 0.50            | 0.40    |
| Mineral premix <sup>a</sup> | 1.00            | 1.00    |
| Vitamin premix <sup>b</sup> | 1.00            | 1.00    |
| Total                       | 1000            | 1000    |
| Nutrients calculated (g/kg) |                 |         |
| ME (kcal/kg)                | 3000            | 3100    |
| Crude protein               | 215.00          | 210.00  |
| Non-phytate phosphorus      | 5.00            | 4.00    |
| Calcium                     | 10.00           | 9.00    |
| Lysine                      | 12.00           | 11.00   |
| Methionine                  | 5.500           | 4.70    |
| Sulfur amino acid           | 9.00            | 8.00    |
| Threonine                   | 8.50            | 8.00    |

<sup>a</sup>Content per kg premix: Manganese – 10,000 mg; Zinc – 7500 mg; Copper – 800 mg; Iron – 10,000 mg; Iodine – 45 mg; Selenium – 44.7 mg.

<sup>b</sup>Content per kg premix: Vitamin A – 1,100,000 IU; Vitamin D – 3,360,000 IU; Vitamin E – 3600 IU; Vitamin K – 3260 mg; Thiamine hydrochloride – 260 mg; Riboflavin – 960 mg; DL-Calcium pantothenate – 1300 mg; Nicotinic acid – 8000 mg; Sinkalin – 66,000 mg; Vitamin B1 – 22.7 mg; Folic acid – 120 mg; Pyridoxine hydrochloride – 530 mg; Biotin – 24 mg.

CP: Crude protein; ME: Metabolizable energy.

as growth promoter dosage/kg diet (contained 300 mg Amoxicillin/kg; Henan Muxiang Veterinary Pharmaceutical Co., Ltd, China) + infected; T2 = birds on 0.75 g activated *S. cerevisiae* ( $1 \times 10^{10}$  cells/g, produced by Lesaffre, France) as probiotic/kg diet + infected; T3 = birds on 0.75 g inactivated *S. cerevisiae* ( $1 \times 10^{10}$  cells/g, produced by Thepax, Milan, Italy) as prebiotic/kg diet + infected. Inactivated *S. cerevisiae* contained 28.90% crude protein, 2.57% lipid, 7.8% ash, 16.00% mannanoligosaccharide, and 18.00%  $\beta$ -glucan. According to the product manufacturer, all dietary supplements of broilers were added by direct mixing after they had been gradually mixed with some feed ingredients to create a premix. At 20 and 21 days old, all birds in the positive control and dietary supplement groups (T1–T3) were orally injected (1 mL) with a laboratory-prepared bacterial suspension of *C. perfringens* (type A) at a concentration of  $1 \times 10^9$  colony-forming units/mL, as described in a previous study by Al-Baadani et al. (2016); Hernandez-Patlan et al. (2019).

### Performance and carcass traits

At 15 and 30 d of age, the body weight (BW) and feed were recorded. Body weight gain (BWG = final

BW – initial BW) and feed intake (FI = feed offered – feed refused) were estimated to calculate the feed conversion ratio ( $FCR = FI/BWG$ ) during the study periods at 1–15 d, 16–30 d and 1–30 d of age, according to previously described methods (Abo Ghanima et al. 2023; Zammit and Park 2023).

At 30 d of age, one bird per cage (12 birds/dietary group) was randomly selected and weighed after 10 h of fasting and before slaughter (pre-slaughter weight). The birds were slaughtered individually under standardised conditions. The carcass weight was measured directly after slaughter to calculate dressing yield as a percentage of pre-slaughter weight (carcass weight/pre-slaughter weight \* 100) according to Naghibi et al. (2023). The carcass parts (wings, breast, saddle, and back) were weighed and expressed as a percentage of carcass weight (carcass parts/carcass weight \* 100) according to Mousapoor et al. (2024).

### Blood sampling and analysis

On day 30, almost 5 mL of whole blood was taken at random from the wing veins of 12 birds per dietary group using a non-anticoagulant tube. The serum was separated after the blood samples were centrifuged at  $3000 \times g$  for 30 min and then stored at  $-80^\circ\text{C}$  until they were analysed for liver enzymes, antioxidant activity and oxidative stress. Alanine aminotransferase (ALT), aspartate aminotransferase (AST), alkaline phosphatase (ALP) and lactate dehydrogenase (LDH) analyzes were performed using a spectrophotometric analyser (Chem Well, FL, USA) and commercially available reagent kits (United Diagnostics Industry, KSA). Total antioxidant capacity (TAC) and malondialdehyde (MDA) in serum were analysed using an ELISA plate reader (MR-96A; Mindray, Shenzhen, China) with an assay procedure to determine TAC and MDA in serum according to the test kit manufacturer's instructions (Cayman Chemical Company, USA), as previously described by Humam et al. (2020).

### Selected caecum microbiota

The caecal contents of 12 birds per dietary group were aseptically collected and stored at  $-20^\circ\text{C}$  for subsequent microbiota analysis. According to the established protocol (Al-Baadani et al. 2023), each sample (10  $\mu\text{L}$ ) was plated on different selective media: *Lactobacillus spp.* was grown on MRS agar and *C. perfringens* on BHI agar and then incubated at  $37^\circ\text{C}$  for 24 h under anaerobic conditions. *Salmonella spp.* and *Escherichia coli* were grown on EMB agar and incubated at  $37^\circ\text{C}$  for 24 h

**Table 2.** Effect of dietary groups on growth performance parameters in broilers.

| Groups <sup>1</sup>                   | Parameters per period |                   |           |                   |                    |                     |                   |                   |                     |
|---------------------------------------|-----------------------|-------------------|-----------|-------------------|--------------------|---------------------|-------------------|-------------------|---------------------|
|                                       | 1 – 15 days           |                   |           | 16 – 30 days      |                    |                     | 1 – 30 days       |                   |                     |
|                                       | BWG (g)               | FI (g)            | FCR (g/g) | BWG (g)           | FI (g)             | FCR (g/g)           | BWG (g)           | FI (g)            | FCR (g/g)           |
| NC                                    | 506                   | 639 <sup>ab</sup> | 1.2600    | 1303 <sup>b</sup> | 1948 <sup>a</sup>  | 1.5000 <sup>c</sup> | 1809 <sup>b</sup> | 2586 <sup>a</sup> | 1.4300 <sup>c</sup> |
| PC                                    | 509                   | 639 <sup>ab</sup> | 1.2600    | 912 <sup>d</sup>  | 1662 <sup>c</sup>  | 1.8300 <sup>a</sup> | 1421 <sup>d</sup> | 2301 <sup>b</sup> | 1.6200 <sup>a</sup> |
| T1                                    | 508                   | 654 <sup>a</sup>  | 1.2900    | 1115 <sup>c</sup> | 1867 <sup>b</sup>  | 1.6800 <sup>b</sup> | 1623 <sup>c</sup> | 2521 <sup>a</sup> | 1.5500 <sup>b</sup> |
| T2                                    | 492                   | 628 <sup>b</sup>  | 1.2800    | 1430 <sup>a</sup> | 1904 <sup>ab</sup> | 1.3300 <sup>d</sup> | 1922 <sup>a</sup> | 2531 <sup>a</sup> | 1.3200 <sup>d</sup> |
| T3                                    | 482                   | 620 <sup>b</sup>  | 1.2900    | 1406 <sup>a</sup> | 1899 <sup>ab</sup> | 1.3500 <sup>d</sup> | 1888 <sup>a</sup> | 2519 <sup>a</sup> | 1.3300 <sup>d</sup> |
| SEM <sup>2</sup>                      | 6.8600                | 7.2200            | 0.0100    | 19.0000           | 23.9000            | 0.0200              | 21.3000           | 28.9000           | 0.0100              |
| Source of variance ( <i>p</i> -value) |                       |                   |           |                   |                    |                     |                   |                   |                     |
| Groups                                | 0.0540                | 0.0110            | 0.0620    | <0.0001           | <0.0001            | <0.0001             | <0.0001           | <0.0001           | <0.0001             |
| PC vs NC                              | 0.7320                | 0.9930            | 0.6190    | <0.0001           | <0.0001            | <0.0001             | <0.0001           | <0.0001           | <0.0001             |
| PC vs T1, T2, and T3                  | 0.6270                | 0.8800            | 0.3120    | <0.0001           | <0.0001            | <0.0001             | <0.0001           | <0.0001           | <0.0001             |

<sup>a–d</sup>Differences between superscript letters for each parameter are significant at  $p < .05$ .

The mean for each dietary group within each parameter includes 12 replicates (cages).

<sup>1</sup>NC = negative control, birds fed control diet + uninfected; PC = positive control, birds fed control diet + infected; T1 = 0.5 g amoxicillin 20%/kg diet + infected; T2 = 0.75 g activated *S. cerevisiae* ( $1 \times 10^{10}$  cells/g) as probiotic/kg diet + infected; T3 = birds supplemented with 0.75 g inactivated *S. cerevisiae* ( $1 \times 10^{10}$  cells/g) as prebiotic /kg diet + infected; BWG = body weight gain; FI = feed intake; FCR = feed conversion ratio..

<sup>2</sup>SEM, standard error of the mean.

under aerobic conditions. The selected caecal microbiota was expressed as log<sub>10</sub> colony-forming units per gram of digesta and calculated using a colony counter.

### Ileal histomorphology

Immediately after slaughter at 30 d of age, approximately 2 cm of ileum tissue was carefully removed from 12 birds per dietary group. The tissue was washed with phosphate buffered saline (PBS), fixed in 10% neutral buffered formalin and processed for histological analysis. After dehydration through graded ethanol, tissues were embedded in paraffin, sectioned at 5 µm thickness using a microtome (Leica, RM 2255, Germany) and mounted on glass slides at 50 °C in a water bath. Sections were stained with haematoxylin and eosin for histologic measurements (villus height, crypt height, and goblet cells) using a light microscope equipped with an image analyser (Nikon, Corp, Japan). The villus surface area and the ratio between villus length and crypt height were calculated as described by Alyileili et al. (2020).

### Statistical analysis

A normality test for all data was performed prior to analysis. The general linear model for a completely randomised design was used to analyse the data. Statistical Analysis System 9.4 (SAS Institute, 2008) software was used to statistically analyse the collected data. Duncan's multiple range tests ( $p \leq 0.05$ ) were used to assess statistical differences between means. Comparisons of means for each parameter were performed by orthogonal contrasts to determine the significant effect of birds infected with *C. perfringens* on a constant control diet without any supplements (PC vs. NC) and the significant effect of

supplements under constant infection conditions (PC vs. T1, T2, and T3). The means of dietary group per parameter was expressed with standard error of the mean ( $\pm$  SEM).

## Results

### Performance measurements

The effects of dietary groups on growth performance parameters are presented in Table 2. The result shows that the performance of the birds before infection with *C. perfringens* (1–15 d) was not affected by dietary groups ( $p > .05$ ), with the exception of FI, which decreased in T2 and T3 ( $p < .05$ ). At 16–30 d and 1–30 d, birds infected with *C. perfringens* fed the basal diet (PC) showed a decrease in BWG and FI compared to NC with a simultaneous increase in FCR ( $p < .05$ ). Whereas, birds fed T2 and T3 followed by T1 showed significantly higher BWG compared to the control groups (NC and PC;  $p < .05$ ). In addition, FI increased and FCR improved in dietary supplemented (T1–T3) compared to the PC ( $p < .05$ ), especially during the 16–30 d period.

### Carcase traits

The effects of dietary groups on carcase traits are presented in Table 3. The result shows that the pre-slaughter weight and carcase weight of birds infected with *C. perfringens* (PC) decreased compared to NC group, while the weight increased again as a result of other dietary groups (T1–T3;  $p < 0.05$ ). The dressing yield and relative weight of carcase components (Wings, breasts, saddle, and backs) did not show any significant changes by infection or dietary supplemented ( $p > 0.05$ ).



**Table 3.** Effect of dietary groups on carcass traits in broilers.

| Groups <sup>1</sup>          | Parameters               |                    |                    |           |             |            |           |
|------------------------------|--------------------------|--------------------|--------------------|-----------|-------------|------------|-----------|
|                              | Pre-slaughter weight (g) | Carcass weight (g) | Dressing yield (%) | Wings (%) | Breasts (%) | Saddle (%) | Backs (%) |
| NC                           | 1813 <sup>a</sup>        | 1167 <sup>a</sup>  | 64.3700            | 10.8700   | 39.9500     | 41.2100    | 7.9700    |
| PC                           | 1502 <sup>b</sup>        | 972 <sup>b</sup>   | 64.7600            | 11.4600   | 39.3700     | 40.9800    | 8.1900    |
| T1                           | 1785 <sup>a</sup>        | 1156 <sup>a</sup>  | 64.7600            | 11.0100   | 39.7500     | 41.4400    | 7.8100    |
| T2                           | 1837 <sup>a</sup>        | 1185 <sup>a</sup>  | 64.5400            | 10.7100   | 40.1800     | 40.9300    | 8.1900    |
| T3                           | 1816 <sup>a</sup>        | 1174 <sup>a</sup>  | 64.6500            | 10.8300   | 39.6700     | 41.4400    | 8.0600    |
| SEM <sup>2</sup>             | 28.0000                  | 19.5000            | 0.5200             | 0.5600    | 0.3600      | 0.3000     | 0.2100    |
| Source of variance (p-value) |                          |                    |                    |           |             |            |           |
| Groups                       | <0.0001                  | <0.0001            | 0.0770             | 0.0320    | 0.5870      | 0.6120     | 0.7230    |
| PC vs NC                     | <0.0001                  | <0.0001            | 0.6140             | 0.0170    | 0.2630      | 0.5710     | 0.4930    |
| PC vs T1, T2, and T3         | 0.0030                   | 0.0100             | 0.6220             | 0.3300    | 0.6660      | 0.7670     | 0.7370    |

<sup>a,b</sup>Differences between superscript letters for each parameter are significant at  $p < .05$ .

The mean for each dietary group within each parameter includes 12 replicates (birds).

<sup>1</sup>NC=negative control, birds fed control diet + uninfected; PC=positive control, birds fed control diet + infected; T1 = 0.5 g amoxicillin 20%/kg diet + infected; T2 = 0.75 g activated *S. cerevisiae* ( $1 \times 10^{10}$  cells/g) as probiotic / kg diet + infected; T3 = birds supplemented with 0.75 g inactivated *S. cerevisiae* ( $1 \times 10^{10}$  cells/g) as prebiotic /kg diet + infected.

<sup>2</sup>SEM, standard error of the mean.

**Table 4.** Effect of dietary groups on liver enzymes and oxidative activity in broilers.

| Groups <sup>1</sup>          | Parameters      |                   |                   |           |                     |                     |
|------------------------------|-----------------|-------------------|-------------------|-----------|---------------------|---------------------|
|                              | ALT (U/L)       | AST (U/L)         | ALP (U/L)         | LDH (U/L) | TAC ( $\mu$ M/L)    | MDA ( $\mu$ M/L)    |
| NC                           | 7 <sup>c</sup>  | 199 <sup>c</sup>  | 888 <sup>d</sup>  | 1188      | 7.5700 <sup>a</sup> | 6.0700 <sup>c</sup> |
| PC                           | 10 <sup>a</sup> | 244 <sup>a</sup>  | 1385 <sup>a</sup> | 1578      | 4.8600 <sup>c</sup> | 9.5500 <sup>a</sup> |
| T1                           | 9 <sup>b</sup>  | 222 <sup>b</sup>  | 1050 <sup>b</sup> | 1147      | 4.5300 <sup>c</sup> | 9.2400 <sup>a</sup> |
| T2                           | 9 <sup>b</sup>  | 238 <sup>ab</sup> | 1201 <sup>b</sup> | 1249      | 5.3200 <sup>b</sup> | 7.1800 <sup>b</sup> |
| T3                           | 8 <sup>bc</sup> | 222 <sup>b</sup>  | 825 <sup>d</sup>  | 1205      | 5.5300 <sup>b</sup> | 7.3300 <sup>b</sup> |
| SEM <sup>2</sup>             | 0.7600          | 9.8200            | 112               | 143       | 0.4500              | 0.7200              |
| Source of variance (p value) |                 |                   |                   |           |                     |                     |
| Groups                       | 0.0070          | 0.0260            | 0.0120            | 0.1530    | 0.0010              | 0.0040              |
| PC vs NC                     | 0.0010          | 0.0020            | 0.0050            | 0.0410    | 0.0010              | 0.0010              |
| PC vs T1, T2, and T3         | 0.0010          | 0.0030            | 0.0210            | 0.3640    | <0.0001             | 0.0020              |

<sup>a-d</sup>Differences between superscript letters for each parameter are significant at  $p < .05$ .

The mean for each dietary group within each parameter includes 12 replicates (birds).

<sup>1</sup>NC=negative control, birds fed control diet + uninfected; PC=positive control, birds fed control diet + infected; T1 = 0.5 g amoxicillin 20%/kg diet + infected; T2 = 0.75 g activated *S. cerevisiae* ( $1 \times 10^{10}$  cells/g) as probiotic/kg diet + infected; T3 = birds supplemented with 0.75 g inactivated *S. cerevisiae* ( $1 \times 10^{10}$  cells/g) as prebiotic /kg diet + infected; ALT = alanine aminotransferase; AST = aspartate aminotransferase; ALP = alkaline phosphatase; LDH = lactate dehydrogenase; TAC = total antioxidant capacity; MDA = malondialdehyde.

<sup>2</sup>SEM, standard error of the mean.

### Liver enzymes and oxidation statues

The effects of the dietary groups on liver enzymes and oxidative activity are presented in Table 4. The result shows that liver enzyme levels including ALT, AST and ALP increased in the PC group compared to the NC group, while the levels of these enzymes decreased with dietary supplements ( $p < 0.05$ ). The LDH level did not show any significant changes by infection or dietary supplemented ( $p > .05$ ). The serum TAC level of birds infected with *C. perfringens* (PC) or receiving T1 decreased compared to the NC group, while it increased in birds receiving T2 and T3 compared to the PC or T1 group ( $p < .05$ ). On the other hand, birds in the PC and T1 groups had the lowest MDA level compared to the NC, while the level decreased in birds receiving T2 and T3 compared to the PC and T1 ( $p < .05$ ).

### Selected microbiota in the caecum

The effects of dietary groups on selected microbiota in the caecum are presented in Table 5. The current result shows that birds receiving PC had the lowest content of *Lactobacillus spp.* compared to the NC group, while it increased in birds receiving T2 and T3 followed by T1 compared to the PC group ( $p < .05$ ). The *Escherichia coli* content did not show any significant changes by infection or dietary supplemented ( $p > .05$ ). The content of *C. perfringens* and *Salmonella spp.* in the caecum of birds infected with *C. perfringens* (PC) were the highest compared to the NC group, while the content of these bacteria decreased in birds receiving T2 and T3 followed by T1 compared to the PC group ( $p < .05$ ).

### Ileal histomorphometry

The effects of the dietary groups on ileal histomorphometry are presented in Table 6. The current result

**Table 5.** Effect of dietary groups on selected microbiota in the caecum in broilers.

| Groups <sup>1</sup>                  | Parameters (log <sub>10</sub> colony-forming units/1 g digesta) |                       |                         |                        |
|--------------------------------------|-----------------------------------------------------------------|-----------------------|-------------------------|------------------------|
|                                      | <i>Lactobacillus</i> spp                                        | <i>C. perfringens</i> | <i>Escherichia coli</i> | <i>Salmonella</i> spp. |
| NC                                   | 10.1300 <sup>ab</sup>                                           | 9.8000 <sup>d</sup>   | 7.8400                  | 8.1500 <sup>c</sup>    |
| PC                                   | 6.6500 <sup>c</sup>                                             | 16.6100 <sup>a</sup>  | 8.1000                  | 10.5300 <sup>a</sup>   |
| T1                                   | 9.1000 <sup>b</sup>                                             | 13.0100 <sup>b</sup>  | 7.9000                  | 9.0200 <sup>b</sup>    |
| T2                                   | 11.3000 <sup>a</sup>                                            | 11.6700 <sup>c</sup>  | 8.3800                  | 8.0700 <sup>c</sup>    |
| T3                                   | 11.5400 <sup>a</sup>                                            | 11.4800 <sup>c</sup>  | 8.5500                  | 7.9800 <sup>c</sup>    |
| SEM <sup>2</sup>                     | 0.3500                                                          | 0.2600                | 0.2100                  | 0.1900                 |
| Source of variance ( <i>p</i> value) |                                                                 |                       |                         |                        |
| Groups                               | <0.0001                                                         | <0.0001               | 0.0540                  | <0.0001                |
| PC vs NC                             | <0.0001                                                         | <0.0001               | 0.3790                  | <0.0001                |
| PC vs T1, T2, and T3                 | 0.0150                                                          | <0.0001               | 0.2320                  | <0.0001                |

<sup>a-d</sup>Differences between superscript letters for each parameter are significant at *p* < .05.

The mean for each dietary group within each parameter includes 12 replicates (birds).

<sup>1</sup>NC = negative control, birds fed control diet + uninfected; PC = positive control, birds fed control diet + infected; T1 = 0.5 g amoxicillin 20%/kg diet + infected; T2 = 0.75 g activated *S. cerevisiae* ( $1 \times 10^{10}$  cells/g) as probiotic/kg diet + infected; T3 = birds supplemented with 0.75 g inactivated *S. cerevisiae* ( $1 \times 10^{10}$  cells/g) as prebiotic/kg diet + infected.

<sup>2</sup>SEM, standard error of the mean.

**Table 6.** Effect of dietary groups on ileal histomorphometry in broilers.

| Groups <sup>1</sup>                  | Parameters             |                                        |                       |                            | Goblet cells (no) |
|--------------------------------------|------------------------|----------------------------------------|-----------------------|----------------------------|-------------------|
|                                      | Villus length (μm)     | Villus surface area (mm <sup>2</sup> ) | Crypt-depth (μm)      | Villus length/ crypt depth |                   |
| NC                                   | 819.1400 <sup>c</sup>  | 0.3500 <sup>b</sup>                    | 55.9800 <sup>a</sup>  | 14.8800 <sup>d</sup>       | 120 <sup>b</sup>  |
| PC                                   | 576.9500 <sup>d</sup>  | 0.2100 <sup>c</sup>                    | 37.4900 <sup>c</sup>  | 15.6000 <sup>c</sup>       | 85 <sup>d</sup>   |
| T1                                   | 749.6300 <sup>c</sup>  | 0.3200 <sup>b</sup>                    | 51.7400 <sup>ab</sup> | 14.8600 <sup>d</sup>       | 128 <sup>a</sup>  |
| T2                                   | 986.0000 <sup>b</sup>  | 0.4500 <sup>a</sup>                    | 52.0100 <sup>ab</sup> | 19.3200 <sup>ab</sup>      | 125 <sup>b</sup>  |
| T3                                   | 1091.0100 <sup>a</sup> | 0.4900 <sup>a</sup>                    | 51.7100 <sup>ab</sup> | 21.5300 <sup>a</sup>       | 123 <sup>b</sup>  |
| SEM <sup>2</sup>                     | 22.3000                | 0.0100                                 | 1.3200                | 0.5900                     | 1.7800            |
| Source of variance ( <i>p</i> value) |                        |                                        |                       |                            |                   |
| Groups                               | <0.0001                | <0.0001                                | <0.0001               | <0.0001                    | <0.0001           |
| PC vs NC                             | <0.0001                | 0.0010                                 | <0.0001               | 0.3960                     | <0.0001           |
| PC vs T1, T2, and T3                 | 0.0400                 | 0.0180                                 | <0.0001               | 0.0140                     | 0.0001            |

<sup>a-d</sup>Differences between superscript letters for each parameter are significant at *p* < .05.

The mean for each dietary group within each parameter includes 12 replicates (birds).

<sup>1</sup>NC = negative control, birds fed control diet + uninfected; PC = positive control, birds fed control diet + infected; T1 = 0.5 g amoxicillin 20%/kg diet + infected; T2 = 0.75 g activated *S. cerevisiae* ( $1 \times 10^{10}$  cells/g) as probiotic/kg diet + infected; T3 = birds supplemented with 0.75 g inactivated *S. cerevisiae* ( $1 \times 10^{10}$  cells/g) as prebiotic/kg diet + infected.

<sup>2</sup>SEM, standard error of the mean.

shows that the histomorphometric parameters were affected by the infection or dietary supplements (*p* < .05). The birds in the PC group had the lowest histomorphometric values (villus length, villus surface area, crypt depth, villus length/crypt depth and goblet cells) compared to the NC group, while they improved in the birds that received dietary supplements (T1–T3) compared to the PC group.

## Discussion

Birds fed the control diet without infection (NC) showed better growth performance than the group of infected birds (PC), which showed a negative effect on necrotic enteritis due to *C. perfringens* infection. This is consistent with the study by Wang et al. (2021), who reported that a reduction in BWG, FI, and increased FCR are typical signs of necrotic enteritis. Birds infected with *C. perfringens*-fed supplements (T1–T3) showed improved growth performance with increased

BWG and FI while decreased FCR compared to the PC group. This improvement in body weight and feed efficiency in infected birds fed supplements, especially activated and inactivated *S. cerevisiae*, may be attributed to several mechanisms, including modulation of the gut microbiota, improvement of digestion and nutrient absorption, and improvement of gut barrier function. By addressing these critical factors, these supplements can effectively mitigate the negative effects of *C. perfringens* infection and promote optimal growth in broiler chickens. The effectiveness of the supplements in mitigating the negative effects of infection is consistent with previous studies (Khalique et al. 2020; Eid et al. 2020). These supplements can improve gut health by modulating the gut microbiota, reducing pathogen colonisation, and improving nutrient absorption, leading to improved growth performance (Khalid et al. 2021; Fazelnia et al. 2021). In addition, Johnson et al. (2020) found that dietary supplementation with inactivated *S. cerevisiae* improved

the performance of birds infected with *C. perfringens*. The data also showed that birds infected with *C. perfringens* (PC) had lower pre-slaughter and carcass weights compared to non-infected birds (NC). However, these weights increased significantly in the infected birds as a result of dietary supplementation. The reduction in pre-slaughter and carcass weights in infected birds could be due to necrotic enteritis caused by *C. perfringens*, resulting in poor nutrient absorption, reduced feed intake, and inflammation, with a negative effect on body weight and carcass traits compared to the negative control. This is in line with the findings of Alqhtani et al. (2024), who found that broilers infected with *C. perfringens* showed reduced growth performance and thus negative effects on carcass traits. The ability of dietary supplementation with activated and inactivated *S. cerevisiae* (probiotics and prebiotics) to improve carcass traits in infected birds is encouraging. A healthier gut environment, reduced inflammation, and improved feed efficiency can contribute to better growth performance and carcass quality, as observed in groups T2 and T3 compared to groups PC and T1. Several studies have reported that dietary supplementation with probiotics and prebiotics can help mitigate the negative effects of *C. perfringens* infection on carcass quality in broiler chickens (Ahiwe et al. 2021; Ugwuoke et al. 2021).

Birds infected with *C. perfringens* (PC) showed elevated liver enzyme levels (ALT, AST and ALP). This indicates that infection with *C. perfringens* can lead to systemic inflammation and liver damage (Ahmed et al. 2023b). The present results are in agreement with those of Alqhtani et al. (2024), who found that the birds in the infected group had higher serum ALT and AST levels than the birds fed a basal diet and were not infected. Conversely, dietary supplementation with activated and inactivated *S. cerevisiae* reduced these enzyme levels. These results suggest that dietary supplements may help to mitigate the detrimental effects of *C. perfringens* infection on liver health (Yalçinkaya et al. 2008; Udeh et al. 2021). In addition, one study showed that the levels of liver function enzymes such as ALT and AST decreased in the serum of birds receiving inactivated *S. cerevisiae* (Abdulrazaq 2023), while another study showed no effect when birds received activated *S. cerevisiae* (Hussein and Selim 2018). Infected birds had lower TAC and higher MDA levels, indicating oxidative stress. The increased oxidative stress indicated by lower TAC and higher MDA levels is also a common finding in necrotic enteritis caused by *C. perfringens* (Hassan et al. 2023). Administration of antibiotics decreased TAC and

increased MDA, while administration of probiotics and prebiotics (T2 and T3) improved antioxidant status in this study. The results of the current study on the effects of dietary supplements on TAC and MDA are consistent with those of many other researchers (Mikelsaar and Zilmer 2009; Capcarova et al. 2010). Mechanisms underlying the beneficial effects of supplements likely include reduced inflammation, improved gut health, and enhanced antioxidant defences. Abdel-Rahman et al. (2013) reported that administration of probiotics lowered MDA levels in broiler chickens. In addition, a study reported that dietary supplementation with mannan oligosaccharides stimulated oxidative defense mechanisms in birds (Ognik and Krauze 2012). The increased activity of TAC could be due to the inhibition of lipid peroxidation by activated and inactivated *S. cerevisiae* (probiotic and prebiotic). Haghighi et al. (2006) concluded that the beneficial bacteria enhance immunity through natural antibodies in chickens. Beneficial bacteria produce some factors that have a chelating effect (Lin et al. 2014). These factors can cause inhibition of ROS, which could potentially prevent oxidative damage and prevent cell membrane damage by increasing antioxidant enzymes (Aluwong et al. 2013).

Birds fed basal diet that were infected (PC) had lower levels of *Lactobacillus spp.* than uninfected birds (NC). This could indicate that an infection with *C. perfringens* can disrupt the intestinal microbiota, creating an environment that favours the growth of pathogenic bacteria. Conversely, dietary supplementation with activated and inactivated *S. cerevisiae* increased the levels of *Lactobacillus spp.* The decrease in *Lactobacillus spp.* and the increase in *C. perfringens* and *Salmonella spp.* in infected birds is consistent with previous studies. Infection with *C. perfringens* can disrupt the gut microbiota, leading to an excessive growth of pathogenic bacteria and a decrease in beneficial bacteria such as *Lactobacillus spp.* (Villagr n-de la Mora et al. 2019). On the other hand, infected birds had higher levels of *C. perfringens* and *Salmonella spp.* than uninfected birds. Whereas, dietary supplementation with activated and inactivated *S. cerevisiae*, decreased these pathogen levels. These results suggest that dietary supplements can help to restore microbial balance and reduce the risk of infection (Mora et al. 2020; Liu et al. 2021). The mechanisms underlying the beneficial effects of dietary supplements with activated and inactivated *S. cerevisiae* can help restore microbial balance by competing with pathogens for resources, producing antimicrobial substances, and modulating the immune response. The improved composition of the gut microbiota can lead to improved



nutrient absorption, reduced inflammation, and improved overall gut health. Birds infected with *C. perfringens* (PC) showed lower ileal histomorphometric parameters (villus length, villus surface area, crypt depth, villus length/crypt depth, and goblet cells) compared to uninfected birds (NC). Dietary supplementation improved these parameters in the infected birds. Infection with *C. perfringens* can cause necrotic enteritis leading to intestinal inflammation, villus atrophy and crypt hyperplasia (Al-Baadani et al. 2016). These changes can impair nutrient absorption and lead to growth retardation (Feng et al. 2010). The ability of dietary supplements to improve ileal histomorphometry in infected birds is encouraging. This suggests that supplements may help to restore intestinal integrity and improve nutrient absorption (Hussein et al. 2020).

## Conclusions

In conclusion, the results showed that activated and inactivated *S. cerevisiae* (probiotics and prebiotics) can effectively mitigate the negative effects of *C. perfringens* infection on broiler performance and health by improving growth performance and carcass weight and reducing liver enzyme levels and oxidative stress markers in infected birds. In addition, positive effects on gut health were observed through an increase in *Lactobacillus spp*, a reduction in pathogenic bacteria and an improvement in ileal morphology in infected birds. However, these results highlight the potential of nutritional interventions to address the challenges associated with *C. perfringens* infections in poultry production. Further re-search is needed to investigate the specific mechanisms underlying these positive effects in broiler chickens.

## Ethical approval

The current study was approved by the Scientific Research Ethics Committee of the King Saud University Research Council (SE-21-47) and follows the ARRIVE guidelines for animal experimentation.

## Authors contributions

The original written paper, the formal analysis, the visualisation, and the conception were prepared by A.S. Alharthi and H.H. Al-Baadani. The methodology, data curation, investigation, review writing, and editing were undertaken by A. Saleh, A.A. Qasem, N.I. Abbas and M.A. Ibraheem. The final paper was reviewed and approved by all authors.

## Disclosure statement

No potential conflict of interest was reported by the author(s).

## Funding

The authors gratefully acknowledge the support of the Researchers Supporting project number (RSPD2025R1109), King Saud University, Riyadh, Saudi Arabia.

## Data availability statement

The data that support the findings of this study are available from the corresponding author (H.H. Al-Baadani), upon reasonable request.

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